

Postura - Step 8: Results Screen

Developer instructions – March 2026

1 PURPOSE

The results screen is what the employee sees immediately after photo capture and ROSA scoring. It must communicate risk clearly without clinical jargon, show what was measured, and give one clear action. This is the primary output of the entire assessment.

Literature confirms: employees who do not understand their result do not act on it. Ergonomics Help (2021): 'If they do not understand why it is better to place the mouse closer, they will not implement it.' Plain language and body region visuals are the two most effective communication formats for ergonomic results.

2 CURRENT SCREEN — WHAT TO KEEP AND WHAT TO CHANGE

Element	Current	Action	Why
Screen title	ISO Ergonomic Analysis	Change text to: ROSA Ergonomic Analysis	ISO is not the method used
Radial dial with %	17% Compliance — 0–100 radial	Remove entirely	% compliance is clinically meaningless to employee. Literature: creates confusion without behavior change.
6 deviation rows	Neck Angle 15 degrees, Wrist 15 degrees etc.	Replace with ROSA sub-score rows — see Section 4 for exact rows, variable names and labels	Degree deviations are not understandable. ROSA component scores are.
Risk by Body Region grid	Elbows, Shoulder, Wrist, Lower Back, Upper Back	Keep exactly — no changes	Best element on screen. Literature confirms body region visualization is most effective format.
Export button	Export ISO Report (PDF)	Rename: Export ROSA Report (PDF)	Label must match method
Back / View Corrections	Existing	Keep exactly	Correct and functional

3 NEW TOP BLOCK — RISK TIER + ROSA SCORE

Replace the ISO radial dial with two elements: a prominent tier label and the ROSA score as supporting detail. Primary message first — supporting data second. This follows health communication best practice.

3.1 The four tiers — what to display

GREEN - Prevention

Your workstation is low risk. Keep your current setup. Reassess in 12 months.

ORANGE - Monitor

Your workstation has risk factors. We recommend adjustments before symptoms develop. Reassess in 3 months.

RED - Acute

Your workstation needs immediate attention. Your employer has been notified.

YELLOW - Refer out

Your workstation looks fine, but you are reporting significant pain. We recommend speaking to a physiotherapist.

YELLOW tier is clinically unique — no existing ROSA tool outputs this. It requires combining ROSA + VAS. Low ROSA + high VAS = workstation is not the problem. Prescribing ergonomic equipment in this case would be clinically wrong and a waste of employer budget. This is a genuine differentiator vs Tumeke and all other ROSA tools.

3.2 ROSA score display below the tier badge

Below the tier badge, show the ROSA score as supporting detail in smaller text:

Field	Value	Style
ROSA Score	7 / 10	Medium size, same color as tier
Label	High risk – immediate action required	Small muted text
Action level	Level 3 – Action required as soon as possible	Small muted text, from Cornell ROSA protocol

4 ROSA SUB-SCORE ROWS — REPLACE ISO DEVIATION ROWS

Replace the 6 ISO deviation rows with 5 ROSA component rows. Each shows what the camera measured for that body area. Labels are plain language — no degrees, no ISO references.

Row	Component	Score shown	Plain label examples	Icon color
1	Chair	3 / 3	Too high / Too low / Optimal	Red/Orange/Green
2	Monitor	2 / 3	Looking down / Looking up / Optimal	Orange/Red/Green
3	Keyboard	1 / 3	Wrists extended / Optimal	Orange/Green
4	Mouse	2 / 3	Reaching / Optimal	Orange/Green
5	Final ROSA	7 / 10	High risk	Red

Source: Technical University of Valencia ROSA software and Tumeke both show component scores with plain labels. Cornell ROSA 2011 confirms: 'Priority actions can be determined based on the partial scores for each workstation element.' Sub-scores are not just display — they tell the employee and employer exactly which element to fix first.

Dart variable names and ranges — use these exactly:

```
// All scores come from the ROSA scoring engine (Step 4 / Mawais) // Use these hardcoded test
values until pipeline is connected: int chairScore = 3; // range 1-3 (1=optimal, 2=borderline,
3=high risk) int monitorScore = 2; // range 1-3 int keyboardScore = 1; // range 1-3 int
mouseScore = 2; // range 1-3 int finalRosa = 7; // range 1-10 (Cornell ROSA 2011 total score)
int maxVasScore = 5; // range 0-10 (highest VAS pain across all body parts) // Plain label
logic per sub-score: // Chair: 1 = 'Optimal' 2 = 'Needs attention' 3 = 'High risk' // Monitor:
1 = 'Optimal' 2 = 'Adjust height' 3 = 'High risk' // Keyboard: 1 = 'Optimal' 2 = 'Wrists
extended' 3 = 'High risk' // Mouse: 1 = 'Optimal' 2 = 'Reaching too far' 3 = 'High risk' //
Icon color per score: // 1 = Green 2 = Orange 3 = Red // Final ROSA tier from
getRiskTier(finalRosa, maxVasScore): // GREEN / ORANGE / RED / YELLOW
```

5 TIER MATRIX LOGIC — HOW TO CALCULATE THE TIER

After ROSA score is computed and VAS scores are collected, run the matrix function. VAS high = any body part reporting pain intensity ≥ 7 out of 10.

	ROSA 1-3 (low)	ROSA 4-5 (medium)	ROSA 6-10 (high)
VAS 0-3 (low pain)	GREEN Prevention	ORANGE Monitor	ORANGE Monitor
VAS 4-6 (medium pain)	ORANGE Monitor	ORANGE Monitor	RED Acute
VAS 7-10 (high pain)	YELLOW Refer out	RED Acute	RED Acute

Dart implementation — matrix function:

```
String getRiskTier(int rosaScore, int maxVasScore) { // maxVasScore = highest pain intensity
reported across all body parts if (rosaScore  $\geq$  6 && maxVasScore  $\geq$  4) return 'RED'; if
(rosaScore  $\leq$  3 && maxVasScore  $\geq$  7) return 'YELLOW'; if (rosaScore  $\geq$  4 || maxVasScore  $\geq$  4)
return 'ORANGE'; return 'GREEN'; }
```

6 RISK BY BODY REGION GRID — KEEP EXACTLY AS IS

The existing Risk by Body Region grid (Elbows, Shoulder, Wrist, Lower Back, Upper Back) is already combining posture deviation with reported pain by body part. This is the most user-friendly element on the screen and is backed by literature.

IAARC 2017 automated ergonomic assessment paper: body segment risk visualization is the most effective format for communicating ergonomic results to non-specialists. The grid format with icons is optimal — employees understand body part icons immediately without needing to read text. Do not change this section.

6b REPLACE: MINI ISO CORRECTION REPORT → ROSA POSTURE GUIDANCE

The current 'Mini ISO Correction Report' screen shows absolute distance measurements ('Lower monitor by 8cm', 'Move keyboard 5cm closer'). These CANNOT be measured from a single side profile photo — MediaPipe has no absolute distance reference. Replace every measurement-based instruction with a directional ROSA-based instruction.

Current text (REMOVE)	Replace with (ROSA-based)	Triggered by
-----------------------	---------------------------	--------------

Lower monitor by 8 cm	Raise your monitor so the top of the screen is at or slightly below eye level.	Monitor score 2 or 3 (neck extension or flexion detected)
Move keyboard 5 cm closer	Move your keyboard closer so your elbows stay at 90° and your wrists stay flat.	Keyboard score 2 (wrist extension or arm reach detected)
Move mouse closer / further	Keep your mouse within easy reach — your elbow should stay close to your body.	Mouse score 2 (lateral arm reach detected)
Raise chair height	Adjust your chair so your knees are at 90° and your feet rest flat on the floor.	Chair score 2 or 3 (knee angle outside 80–100°)
Lower chair height	Lower your chair so your thighs are parallel to the floor and knees at 90°.	Chair score 2 (knees above hip level detected)
Adjust backrest	Sit back fully against your backrest so your lower back is supported.	Backrest score 2 (trunk angle >28° or hip forward of backrest)

Rule: show only the instructions that are triggered by the user's actual ROSA sub-scores. Do NOT show all instructions every time. If chair score = 1, do not show the chair instruction. Maximum 3 instructions visible at once — show the highest-scoring issues first.

6c EQUIPMENT RECOMMENDATION TEXT — OWN WORDS BASED ON ISO 9241-5:2024

Replace all absolute measurements (8–12cm, 10–15cm, +3cm) in the equipment screen with plain-language descriptions based on ISO 9241-5:2024 principles. No ISO text is copied verbatim — all wording is original. Cite 'Based on ISO 9241-5:2024 and Cornell Ergo' as the source line.

Equipment item	Current text (REMOVE)	Replace with — own words	ISO 9241-5:2024 principle
Adjustable monitor riser	Height-adjustable stand (8–12 cm range) to position monitor at eye level	A height-adjustable monitor stand that lets you position the top of your screen at or just below eye level when sitting upright.	Visual display units shall be positioned so the operator can view the screen with the head in a natural, upright position. ISO 9241-5:2024 §6.3
Adjustable footrest	Angled footrest with height adjustment (10–15 cm)	A stable, height-adjustable footrest that supports your feet flat when the desk height prevents full floor contact.	The feet shall be supported on the floor or a footrest. ISO 9241-5:2024 §6.2
Armrest padding / risers	Adjustable armrest pads (+3 cm height)	Padded armrest supports that raise or cushion the arm contact point so your shoulders stay relaxed and elbows rest at desk height without strain.	Armrests shall support the forearms without causing shoulder elevation or restricting movement. ISO 9241-5:2024 §6.2
Lumbar support cushion	Memory foam lumbar roll for lower back support	A shaped lumbar support that maintains the natural inward curve of your lower back when sitting fully back in the chair.	The backrest shall support the lumbar region of the spine. ISO 9241-5:2024 §6.2

Document holder	(if present)	A document holder positioned at the same height and distance as your screen so you can read without repeatedly tilting your head.	Source documents shall be positioned to avoid repeated head movements. ISO 9241-5:2024 §6.4
Keyboard wrist rest	(if present)	A padded wrist rest that supports your wrists in a neutral flat position during pauses in typing — not during active keystrokes.	The working surface shall allow a comfortable forearm and wrist position. ISO 9241-5:2024 §6.2

Source line to show in app under each equipment item: 'Based on ISO 9241-5:2024 and Cornell Ergo'. Do NOT show ISO page numbers or section references to the user — only to the CPE review screen. Keep the High Priority / Low Priority badges and +40% improvement indicators as they are.

7 FINAL SCREEN LAYOUT — TOP TO BOTTOM

#	Block	Change?	Source
1	Screen title: ROSA Ergonomic Analysis	Text change only	Cornell ROSA 2011
2	Tier badge: GREEN/ORANGE/RED/YELLOW + plain language sentence	NEW — replaces ISO circle	ROSA + VAS matrix — novel combination
3	ROSA score: X/10 + action level label	NEW — replaces % compliance	Cornell ROSA 5 action levels
4	5 ROSA sub-score rows with plain labels	Replace ISO deviation rows — see Section 4 for exact detail	Cornell ROSA component scores
5	Photo thumbnail	Keep exactly	Existing
6	Risk by Body Region grid (5 body parts)	Keep exactly — no changes	Existing — best element
7	Export ROSA Report (PDF) button	Rename only	Existing
8	Back / View Corrections buttons	Keep exactly	Existing

8 WHAT MAKES THIS SCREEN UNIQUE vs ALL OTHER ROSA TOOLS

Feature	Postura	Tumeke	Cornell / Valencia ROSA
ROSA sub-scores shown	Yes	Yes	Yes
Plain language labels	Yes	Partial	No
Body region risk grid	Yes	No	No
Combined ROSA + VAS tier	Yes – GREEN/ORANGE/RED/YELLOW	No	No
YELLOW – refer out tier	Yes	No	No
Employee-facing screen	Yes	No – assessor only	No – paper form

MSD prevention claim	Yes – ORANGE catches risk before pain	No	No
----------------------	---------------------------------------	----	----

9 DEV TIME ESTIMATE

Exercise screen and equipment recommendation screen are already built in the app. Step 8 only replaces the top section of the results screen (ISO circle → ROSA tier badge) and wires the ROSA final score as the new trigger input into the existing engines. Body region grid stays exactly as is. No exercise or equipment logic changes needed.

Task	Time
Remove ISO radial dial + % compliance block	30 min
Build tier badge component (4 color states)	45 min
Build ROSA score + action level display	30 min
Replace 6 ISO rows with 5 ROSA sub-score rows	30 min
Implement getRiskTier() matrix function	30 min
Wire ROSA score as trigger input into existing exercise + equipment engines	30 min
Rename Export button	5 min
Keep body region grid, exercise screen, equipment screen exactly as is	0 min — no changes
Total Step 8	~1.5–2 hours (exercise + equipment screens already built)

Sources

1. Cornell ROSA 2011 — 5 action levels — ergo.human.cornell.edu/ROSA-Instructions-2011-2012.pdf
2. Technical University of Valencia ROSA software — action level table — ergonautas.upv.es/rosa
3. IAARC 2017 — body segment visualization most effective for non-specialists — iaarc.org/ISARC2017-Paper052.pdf
4. Ergonomics Help 2021 — employee understanding drives compliance — ergonomicshelp.com/blog/improve-compliance
5. Moriguchi 2022 BMC — ROSA validation — pmc.ncbi.nlm.nih.gov/articles/PMC9160176/
6. PLoS ONE 2019 — combined objective + subjective risk matrix — pmc.ncbi.nlm.nih.gov/articles/PMC6890250/